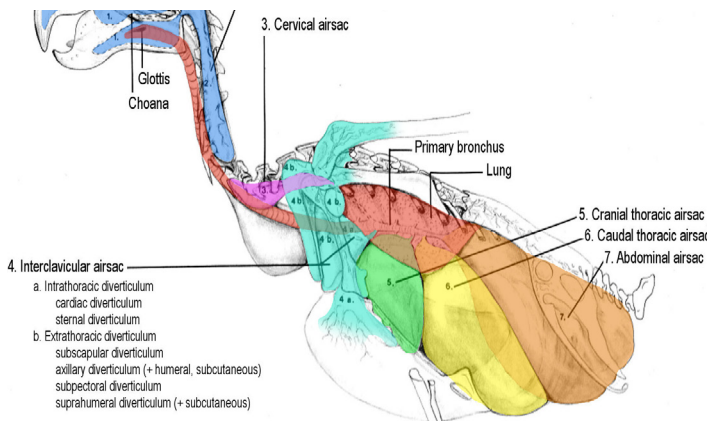


Respiratory Disorders of Birds

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INTRODUCTION

Please refer to the lecture on the anatomy of birds for more background information on anatomy. Knowing the respiratory anatomy of birds is of great relevance when diagnosing and treating diseases of the respiratory system (see refresher below).



The most important features I would definitely recommend you brushing up on are:

- The **infraorbital sinus**: single paranasal sinus of birds that has a soft tissue lateral wall (meaning it is surgically accessible and it can become externally swollen too), has a dorsal opening to the nasal cavity (meaning exudate don't drain well), has several diverticuli including the large cervicocephalic diverticulum, and pneumatizes the maxilla.

- The **trachea** has **complete tracheal rings**, which leads birds to being more prone to trauma and requiring the use of uncuffed endotracheal tube for intubation.

- Sounds are produced in the syrinx**, which means that voice changes are associated with syringeal diseases or extra-syringeal masses compressing the syrinx.

- The lungs are super-duper efficient with a much **higher exchange surface than mammals**, which translates into birds being super-duper sensitive to airborne toxins.

- The air sac system **pneumatizes** most of the body, which means that respiratory infection can easily become systemic. Conversely, systemic diseases often involve the respiratory system. The good thing though is that we can use the air sac system to our advantage for diagnostics (coelioscopy) and treatment (air sac tubes).

This presentation will heavily focus on pet birds, mainly parrots, but other species will also be mentioned when relevant.

CLINICAL EVALUATION

Obviously, birds with respiratory diseases will present with dyspnea, but birds with respiratory lesions don't necessarily present with respiratory signs when mild (air sac aspergillosis for instance). Because birds are good at compensating and hiding clinical signs, they may only show clinical signs when diseases are very advanced and show up as an "acute on chronic" presentation. Since their respiratory system is so efficient, acute respiratory distress from a peracute disease is also common in them. Overall, **diseases that tend to present as acute respiratory distress are tracheal obstruction (foreign bodies, tracheal stenosis), acute and overwhelming infection (acute aspergillosis, bacterial pneumonia), or airborne toxicosis (Teflon).**

Other presenting signs include signs of upper respiratory diseases such as sneezing, coughing, periorbital swelling. Conjunctivitis is also frequently associated with upper respiratory diseases in pet birds. A swelling under the eye is typical for infraorbital sinusitis as well as feather loss around the eyes. When the nostrils are

obstructed, you may notice more the expansion of the infraorbital sinus laterally with each breath. Common signs of lower respiratory diseases include general dyspnea and exercise intolerance.



It is important to **preoxygenate** birds that are dyspneic prior to capture and restraint for a few minutes. If the bird is not deemed stable, then a restricted physical examination and stepwise approach should be followed. A restricted physical examination should be fast and mainly focus on body condition (to have a feel for the chronicity of the disease), coelomic palpation (for extrarespiratory abnormalities compressing the air sacs), examination of the glottis and choana, and cardiopulmonary auscultation. Prior to physical examination, the dyspnea should also be characterized into **inspiratory, expiratory, or mixed**. Inspiratory dyspnea is more common with upper respiratory diseases (including trachea and syrinx) and expiratory and mixed dyspnea are

more common with lower respiratory diseases.

Dyspneic birds are frequently **tail bobbing** and show increased respiratory efforts (increased excursions of the keel). Sometimes, their voice is altered, which is associated with diseases of the syrinx or masses in the thoracic inlet. **When putting birds back in their enclosure, a slower return to normal (>2-3 minutes) may be associated with respiratory conditions.** Open beak breathing is typically a bad sign and may warrant immediate action, such as the placement of an air sac cannula to bypass the trachea and syrinx.

The lungs are auscultated over the dorsum as well as the air sacs. For the air sacs, auscultate caudodorsal just cranial to the tail on either side.

DIAGNOSTIC WORKUP

The diagnostic workup for respiratory diseases is initially similar than for most diseases and include blood work (CBC and biochemistry) and diagnostic imaging (radiographs or CT).

On the CBC, a leucocytosis may be seen with inflammatory disorders such as bacterial and fungal infection. Erythrocytosis may be seen with chronic respiratory disorders in response to chronic hypoxemia (make sure the PCV is not increased because of dehydration). Biochemical findings are also non specific, but an increase in CK (muscle enzyme) may be seen with increased respiratory efforts or lesions causing soft tissue damage (neoplasia and severe infection). Venous blood gases may be helpful to further characterize ventilation disorders. Arterial blood gases are useful to evaluate the degree of hypoxemia, but arterial blood samples are hard to obtain in birds and is typically only performed on large birds. Pulse oximetry may help diagnose and monitor hypoxemia, but absolute values are not reliable in birds as they are calibrated on the mammalian hemoglobin dissociation curve.

In the case of rhinitis and sinusitis, samples can be obtained for cytology and culture. **The infraorbital sinus can easily be accessed laterally between the eye and the beak in a depression that is easily palpated.** It can either be aspirated or fluid can be injected and re-aspirated for cytology and culture. Alternatively, a culture swab may be taken from the choana. We typically don't run culture on nasal swabs because of bacterial contamination with opportunistic bacteria.



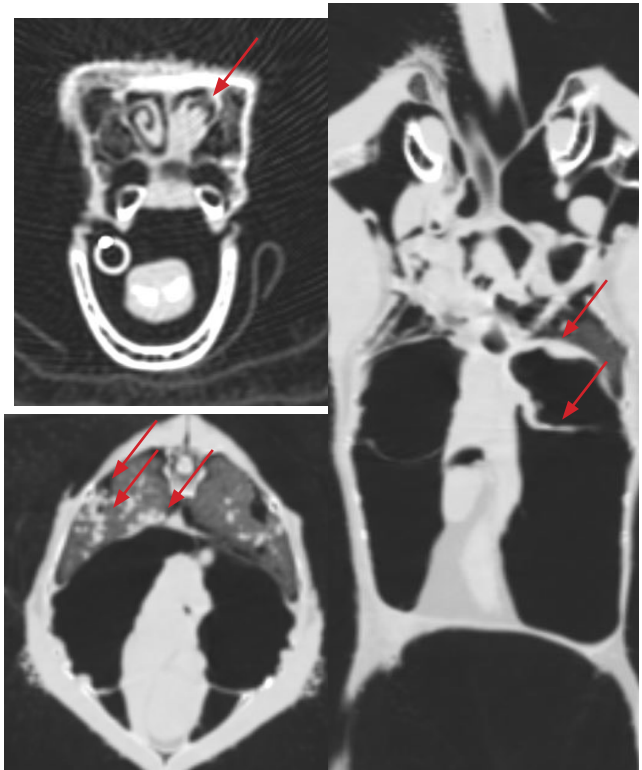
Radiographs are great as the infraorbital sinus – lung – air sac system provides great contrast within the avian body. The lungs have a honeycomb pattern, which when it becomes more prominent is called a **parabronchial pattern**. **The air sac membranes are not normally visible** on plain radiographs. Lesions that are frequently seen are increased opacity of the pulmonary parenchyma (first picture below), thickened air sac membranes (may form “caverns”, second picture below), and granuloma and masses.



Tracheal and syringeal diseases may also be detected on xrays. Upper respiratory diseases are not well visualized on xrays because of the superimposition of anatomical structures.

CT scan is simply the best non-invasive diagnostic im-

aging test one can do to assess the respiratory system of birds. Unfortunately it is not widely available (yet) in private veterinary practices. Lesions in the nasal cavities, infraorbital sinus, trachea, syrinx, bronchi, lungs, and air sacs are readily visualized. Air sac membranes are also visible on CT.



Endoscopy is a bit more invasive but allows direct visualization of the lesions as well as sampling for culture, cytology, and histopathology. Rhinoscopy, choanoscopy, and tracheoscopy are used for upper respiratory diseases. **Coelioscopy through a left or right lateral approach from within the air sacs is used to gain access to the lung – air sac system.** Targeted biopsies of the air sacs and lungs (below picture) can be obtained using 5Fr biopsy forceps in most birds.



GENERAL THERAPY

Oxygen therapy

The goal of oxygen supplementation is to increase the oxygen concentration of inspired air (fraction of inspired O₂: FiO₂), improve blood oxygenation, and increase tissue delivery of O₂. Oxygen therapy is typically **implemented using an oxygen cage** in birds because it is low-stress and noninvasive.



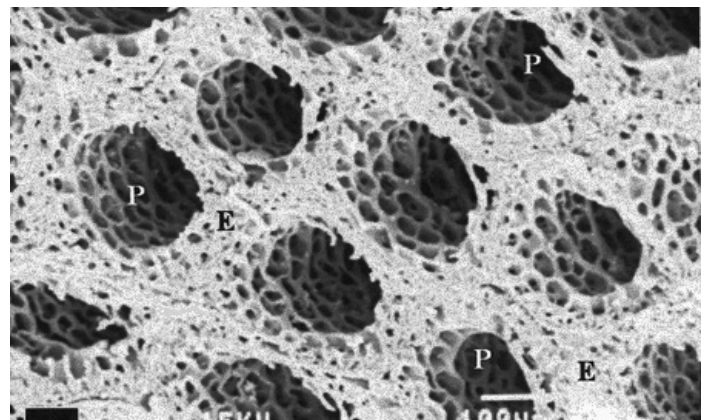
Response to therapy should also be used for monitoring and to titrate the oxygen flow. Oxygen therapy seems extremely beneficial to birds, which have higher metabolic demand for oxygen and may benefit more from increased FiO₂ due to their more efficient gas exchange surface. Sedation with midazolam in dyspneic birds may also reduce anxiety thereby resulting in better ventilation. The response to oxygen therapy depends on the disease (more effective in pulmonary parenchymatous diseases for instance than in obstructive diseases). While the degree of response to oxygen therapy may be variable, regardless of the underlying conditions, **any patient with acute respiratory distress and signs of hypoxia (cyanosis, dyspnea, tachypnea, open-mouth breathing) may benefit from supplemental oxygen.**

The most common complication associated with oxygen therapy is oxygen toxicity because oxygen is a potent oxidizing agent so be careful not to give too high of an oxygen concentration for too long.

Nebulization

The goals of nebulization therapy are to deliver air-borne therapeutics directly to diseased respiratory tissues and to moisten airways in order to increase clearance of exudates and necrotic debris. Local therapeutic delivery may be indicated when vascularization is decreased in diseased tissues (e.g. inflammatory and infected exudates in airways) or in low-perfused respiratory areas such as the air sacs. Nebulization allows to reach high local concentration of therapeutics maximizing its efficacy while minimizing systemic absorption, reducing potential for toxicity, and reducing drug biotransformation.

Depending on particle sizes and airway diameters, aerosolized therapeutics may only reach the upper respiratory system, the larger airways, or lungs and air sacs. Birds' airways have extremely small diameters with parabronchi diameters measuring 0.5-2 mm and air capillaries measuring 3-10 μm .



In comparison, mammalian alveoli have an approximate diameter of 30-150 μm . **Consequently, birds' airways require smaller particle sizes and particle diameters of 0.5- 3 μm are generally recommended.** A pediatric jet nebulizer is the most commonly used nebulizer in avian medicine.



Nebulization time is typically 15-30 minutes and frequency is typically once to twice daily. A variety of drugs can be nebulized such as antibiotics (amikacin, enrofloxacin), antifungals (amphotericin B, clotrimazole), bronchodilators (terbutaline, aminophylline), or just saline (normal saline or hypertonic saline).

AVIAN CHLAMYDIOSIS

Avian chlamydiosis is the main **zoonosis** of pet birds. It is mainly caused by ***Chlamydia psittaci***. This bacteria is tiny and an obligate intracellular bacteria. In young raptors (especially juvenile red-tailed hawks in California), we also have *Chlamydia buteonis* resulting in systemic infection. Lastly, *Chlamydia avium* is another species that has been recovered in parrots and pigeons.

Chlamydia spp. have a biphasic development with 2 forms, an **elementary body** that is the infectious form, and a **reticulate body**, which is the replicating form (and which generates the inclusion bodies in the cell). The disease is contagious and can be transmitted through many routes such as oro-fecal, infectious aerosols, and through body secretions. All birds are susceptible and not all infected birds are sick. Among pet birds, we tend to diagnose this bacteria **mainly in budgerigars and cockatiels** kept in aviaries and in pet stores. It can cause disease outbreak in parrot colonies. *C. psittaci* has several serotypes that tend to be specific to a group of birds. Avian chlamydiosis is mainly a systemic disease with a specific tropism for the respiratory system and liver. As such, it will manifest as rhinitis, conjunctivitis (mainly in budgies, cockatiels, and pigeons), general sickness, pneumonia, air sacculitis, and hepatitis (green urofeces). When transmitted to humans, it causes an atypical pneumonia. While parrots can definitely transmit the disease to humans (hence it is also called psittacosis), most of the human cases have involved other species of birds such as urban wildlife.

Avian chlamydiosis is a reportable disease in California (to the CDEA).

[Official public health guidelines](#) have been developed for the definition of cases and treatment recommendations.

The diagnosis is mainly achieved through a **PCR on an oro-choanal-cloacal swab**, to detect shedding through all the different routes. Serology tests are also available (elementary body agglutination and ELISA) but have a long turn-around time and only diagnose exposure. Confirming the diagnosis requires culturing the bacteria (specific cell, egg culture required) or visual-

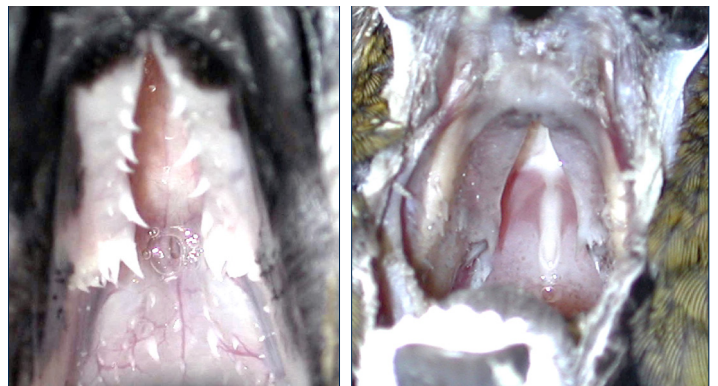
izing the organism on histopathology or cytology. Special histopathology and cytology stains may highlight the organism such as Gimenez or Macchiavello stains. Blood work will typically show a **marked inflammatory leucogram** with or without increase in liver analytes. These changes are non-specific (as most blood work abnormalities). Various changes can be detected on imaging. **The spleen, in particular, is frequently enlarged.**

Regarding treatment, antibiotic with intracellular distribution should be chosen such as tetracyclines and macrolides. **Doxycycline and azithromycin** are the main antibiotics used and it is recommended to **treat for 20-45 days**. As the disease is frequently encountered as an outbreak in colonies of budgerigars or cockatiels, individual treatments are not convenient and specific protocols have been developed for flock treatment. **Cockatiels should be treated with doxycycline in the water whereas budgerigars should be treated with doxycycline supplemented in the feed** (they don't drink much).

NOSTRILS AND NARES

Infectious rhinitis may be caused by a variety of agents. Gram negative bacteria and *Chlamydia psittaci* are commonly reported. Uncomplicated avian chlamydiosis frequently present as only mild rhinitis and conjunctivitis in cockatiels and pigeons.

Chronic granulomatous rhinitis (or rhinolith) is frequent in psittacine birds on a seed only diet and is particularly common in cockatiels, Grey, and Amazon parrots. It seems to be promoted by nutritional deficiencies notably in **vitamin A**. Hypovitaminosis A is also frequently associated with blunting of the choanal papilla (see below, left is normal).



Concretions of desquamated epithelium, debris, necrotic and inflammatory materials occupy the nasal

cavity and induce progressive tissue destruction, which can be extensive. Opportunistic bacterial and sometimes fungal organisms are frequently involved in the disease process.

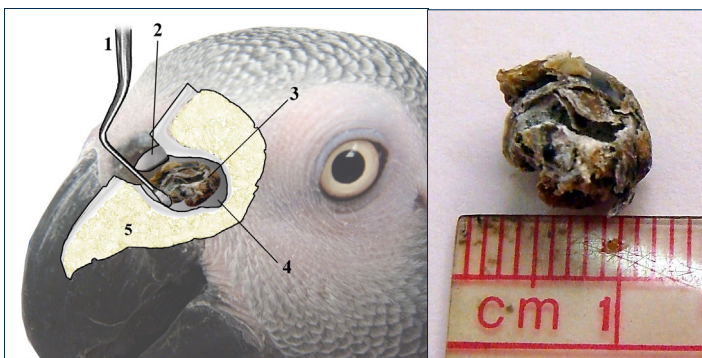
Cnemidocoptic mites are common in budgerigars and may lead to hyperkeratosis of the cere, which ultimately may obstruct the nostrils.

Regarding diagnosis, cultures can be taken in case of rhinitis and skin scraping of the lesion will detect cnemidocoptic mites. Radiographs are typically not very useful for these disorders, but CT scan may help to assess the extension of the rhinitis (and sinusitis).

In terms of treatment, systemic antimicrobial, anti-inflammatories and vitamin A supplementation are frequently used. For rhinitis, **nasal flushing** (with the head upside down) can be performed with saline with or without the addition of antimicrobial and proteolytic enzymes.



Rhinolith have to be surgically debrided. A simple dental or nasal curette is enough to dislodge and extract the nasal plug and then clean the nasal cavity. It should be thoroughly flushed after. The nasal deficit left by the granuloma is often extensive.



Since the integrity and function of the nasal structures are compromised, this procedure will unfortunately have to be repeated periodically to clean nasal debris

that constantly accumulates.

INFRAORBITAL SINUS

Many of the microbial agents that cause rhinitis may also cause sinusitis and both may be seen concurrently. **Periorbital swelling**, feather loss, and erythema are the classical signs.

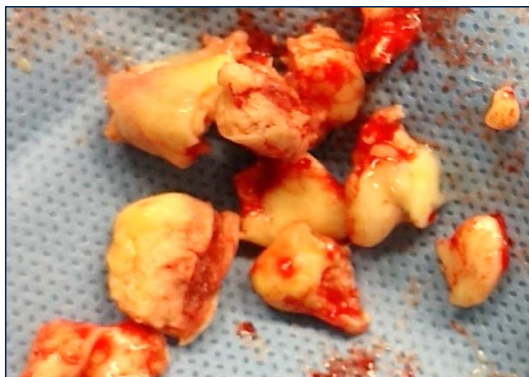


Gram-negative bacteria tend to predominate. Several bacteria deserve further mention because of their importance in poultry or wild birds. Poultry suffer from a high rate of sinusitis for which ***Mycoplasma gallisepticum***, ***Avibacterium paragallinarum*** (infectious coryza of chicken), and ***Bordetella avium*** (turkey coryza) are common etiologic agents. *Mycoplasma gallisepticum* is also a significant disease of **house finches and goldfinches** in North America and present with conjunctivitis and sinusitis and significant morbidity. *Mycoplasma* infection is also associated with upper respiratory disease in psittacine birds. However, *Mycoplasma* spp. appear to be commensal organisms in birds of prey.

Diagnosis is typically achieved through CT-scan of the head and culture/cytology of sinus samples. A sinus lavage or sinus sampling can easily be done by inserting

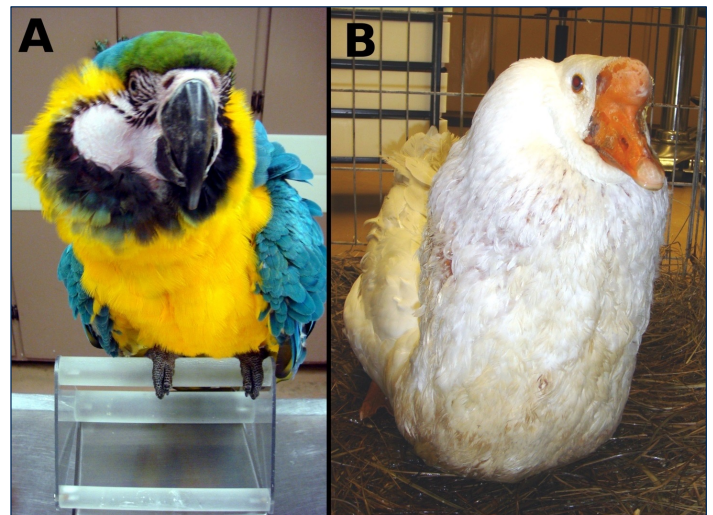
a needle laterally (see picture “Sinus FNA”). Radiographs are not sensitive for these disorders except in severe cases.

The medical treatment for sinusitis depends on the isolated bacterial or fungal pathogen and the amount of exudate present in the sinus. For chronic sinusitis or granulomatous sinusitis, a **sinusotomy** may be attempted. The infraorbital sinus is generally easily approached from just below the eye and jugal arch and this surgery is most commonly done in Galliformes (see below in a peahen).



The term **air sac rupture** is used for any large collection of air subcutaneously. It typically occurs in the neck and is most likely caused by a rupture or overinflation of the **cervicocephalic diverticulum of the infra-**

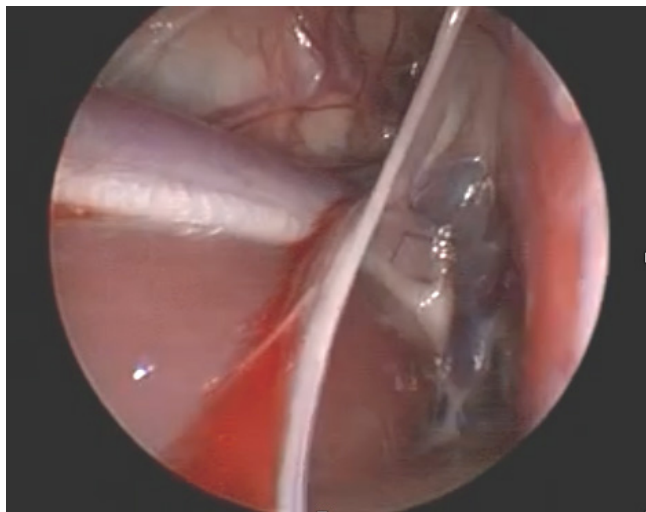
orbital sinus (so not an air sac per se). It can also occur elsewhere in the body following trauma that results in air leakage from an air sac.



The short-term treatment consists of periodic air aspiration with a large syringe and anti-inflammatory and antibiotic treatment (in case of suspected sinusitis). It may resolve on its own after a few days so periodic deflation is definitely an option even if it fills up again. Creating a surgical stoma (marsupialization) or placing a stent may also provide temporary relief, but is rarely effective for more than a few days (which may be enough in some cases) as they tend to close or get plugged.



In chronic cases where these treatments did not work, pleurodesis with a fibrotic agent such as doxycycline may work (it is injected in the air cavity and a bandage is placed around the neck to allow tissue-to-tissue contact). In some cases the perforation can be found by endoscopy (see below) and closed.



TRACHEA AND SYRINX

Tracheal diseases typically result in acute **inspiratory dyspnea** characterized by coughing, wheezing, and open-beak breathing.

Various bacterial agents have been implicated in tracheitis. Viral diseases of the trachea cause diphtheritic lesions and include mainly herpesvirus infections such as the **infectious laryngotracheitis virus in chickens** (and also pheasants and guinea fowls), the Amazon tracheitis virus in Amazon parrots and a few other psittacine species (in recently imported birds), cytomegalovirus in Australian finches, psittacid herpesvirus 3 and 5 infection in South East Asian parrots, and the diphtheric/wet form of poxviruses typically encountered in passerine birds and small psittacines.

Fungal infections of the trachea are seen notably with **tracheal and syringeal aspergillosis**. Several parasites are known to infest the avian trachea and include nematodes such as *Syngamus* spp. (mainly *S. trachea*) and *Cyathostoma* spp. (mainly *C. bronchialis*) and acarids such as *Sternostoma tracheacolum*. **Tracheal worms** infest a wide range of species but clinical diseases are most frequently encountered in captive birds of prey, poultry, and gamebirds. The life cycle of *S. trachea* is direct and earthworms may act as paratenic hosts (larvae may remain viable for up to 3 years in earthworms), which facilitates re-exposure in a closed environment with access to the ground. *C. bronchialis* infests mainly Anseriformes and birds of prey and require earthworms as obligate intermediate hosts. Tracheal worms are treated by using ivermectin and endoscopic removal.



Sternostoma tracheacolum (tracheal, air sac mites) infests primarily canaries and Gouldian finches. The mite has a direct life cycle and is passed from parents to chicks. It is usually treated with oral or percutaneous ivermectin.



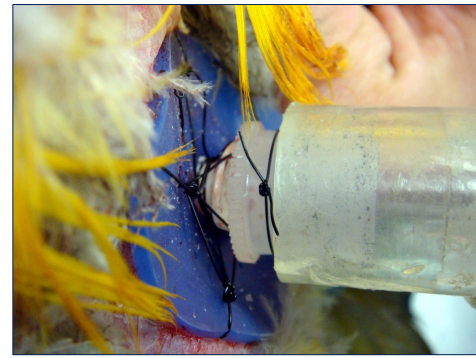
Non-infectious diseases of the trachea include foreign bodies (typically a millet seed), tracheal stenosis, tracheal trauma and collapse, tumors of the glottis or elsewhere, and external occlusions (see later). In particular, **tracheal stenosis is frequently caused by a recent intubation event about 2 weeks prior**. Tracheal trauma are mainly caused by predator attacks in outside birds (chickens, ducks and others). Most treatments are surgical and consist in **tracheotomy or tracheal resection and anastomosis**.



Tracheal foreign bodies can also be extracted using suction in some cases. In case of a seed foreign body in

the trachea of a small bird (too small for tracheoscopic treatment or tracheotomy), a small needle may be inserted just distal to the seed and the seed may be flushed out with air. Alternatively, the seed may be pushed inside the lungs, which is not ideal, but may be a lifesaving procedure. It is not possible to do a tracheotomy in small birds as it would be difficult to close the tracheal incision and it will reduce the lumen significantly post-operatively.

Tracheal diseases can be diagnosed with diagnostic imaging such as radiographs and CT-scan, but **tracheoscopy** is the method of choice to confirm a diagnosis and potentially treat at the same time. Cultures, cytology, and biopsies can be taken endoscopically. Nebulization of drugs may also be used as part of the treatment along with systemic antimicrobial and antiparasitic depending on the underlying cause.



LUNGS AND AIR SACS

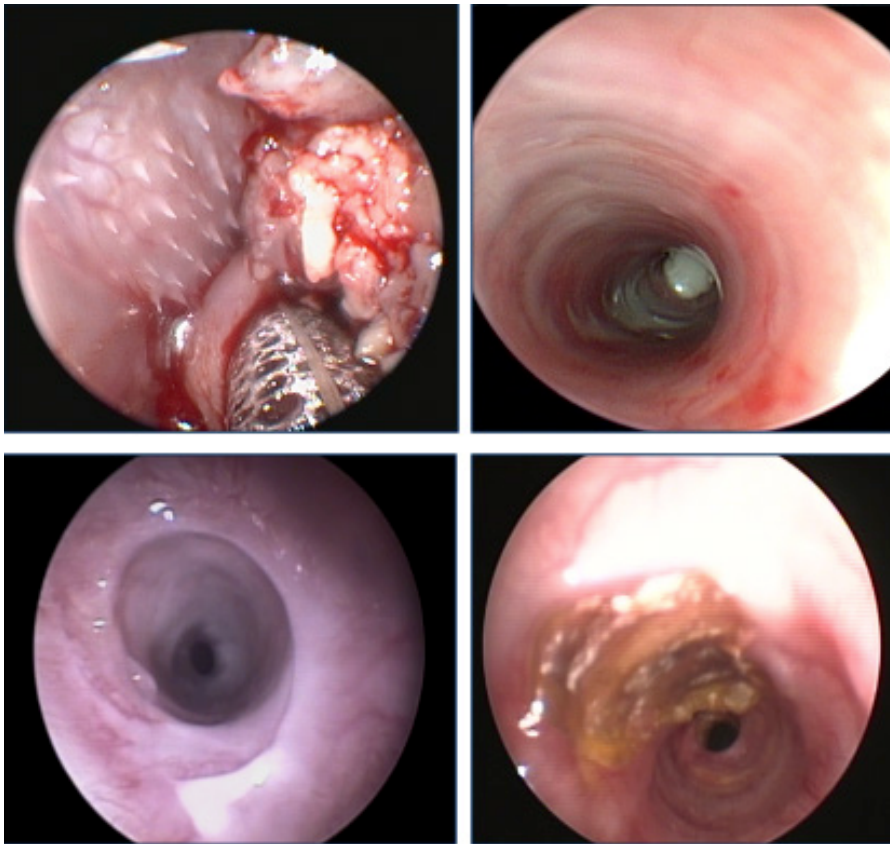
Infectious diseases

Lower respiratory diseases are common in birds and infectious causes tend to predominate. Bacterial pneumonia and air sacculitis can be caused by a variety of bacteria but gram negative bacteria seem to predominate. As mentioned previously, *Chlamydia psittaci* is an important cause of respiratory infection in birds. Chickens also tend to have a variety of bacteria causing upper (see previously) and lower respiratory diseases. Hard-to-treat mixed respiratory infection is extremely common in backyard poultry. Avian tuberculosis caused by *Mycobacterium avium* or *genavense* is mainly a gastrointestinal and liver disease, but may also involve the respiratory system. In chickens, **infectious bronchitis** (adenovirus) is another important respiratory disease to be aware of.

There are several parasites that may infest lungs and air sacs in birds, primarily in wild or zoo birds. These include air sac flukes and **air sac nematodes (filarioid nematodes)**. The tracheal mite can also be found in the air sac.

Of particular interest is the ability in birds to bypass the larynx/trachea/syrinx by placing an **air sac tube in the caudal thoracic or abdominal air sac** on either side. The approach is similar as for a lateral coelioscopy (see later). The tube can only be left for a few days as it causes a lot of internal damage. A filter may be placed to prevent inhalation of foreign materials. An e-collar may be needed to prevent the bird from removing it.

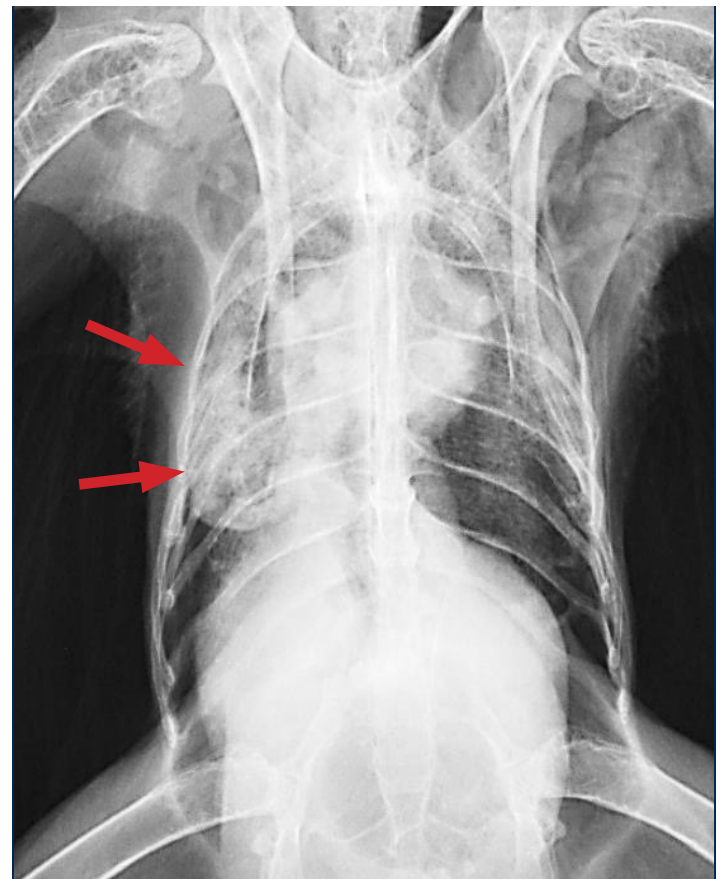
By far, the most important infectious disease of the lung-air sac system is **aspergillosis**. Avian species reported to be more susceptible to respiratory aspergillosis include northern raptorial species (snowy owls, gyrfalcons, rough-legged hawks), juvenile red-tailed hawks, golden eagles, northern goshawks, penguins, flamingos, seabirds (pelicans, auks, gannets, gulls, petrels, loons...), certain parrot species, waterfowl, storks, and mynahs.





Risk factors that have been associated with an increased susceptibility include lead toxicosis (waterfowls, raptors), smoke inhalation injury, captivity (especially seabirds), stress, **steroids**, neoplasia, circovirus infection, hypovitaminosis A, and others. Since the air sac system pneumatizes most of the avian body, aspergillosis lesions may extend to any internal organ by local invasion or distantly by hematogenous dissemination. In addition, *Aspergillus* spp. produce a wide variety of mycotoxins, some with immunosuppressive effects such as the **gliotoxin**, which may participate in the disease progression and pathogenesis. Aspergillosis can be suspected on diagnostic imaging such as radiographs or CT-scan.

Aspergillus fumigatus is the main species isolated but other *Aspergillus* species can be cultured. **Aspergillosis is not zoonotic, is not contagious, and any environmental strain may be potentially pathogenic.** The pathogenesis seems to rather involve either exposure to a **high spore load resulting in acute pulmonary aspergillosis** or some degree of immunosuppression leading to **chronic pulmonary – air sac aspergillosis**. **Tracheal and syringeal aspergillosis** constitutes a third form of the disease (see tracheoscopy pictures).



A **strong inflammatory leucogram** is also frequently, but not always, present on the CBC. The diagnosis can be confirmed on [coelioscopic examination](#) followed by endoscopic biopsies of lesions, histopathology, and/or cytology and fungal culture.

The treatment of aspergillosis is complicated, long, and multifaceted. Usually a combination of nebulized, local, and systemic antifungals is frequently used, but antifungal monotherapy has resulted in good outcomes in some cases (especially with voriconazole). The prognosis should be determined based on radiographic,

computed tomographic, and/or coelioscopic evaluation. **The treatment is lengthy with typically 3-8 months of oral antifungal medications.** Ideally, the treatment should be based on fungal culture and sensitivity, but laboratories offering fungal sensitivity are few and minimum inhibitory concentrations of antifungals for avian fungal isolates may not be available. Available antifungals include the **polyenes** (e.g. **amphotericin B**, not absorbed orally), which bind ergosterol (fungal wall constituent) and permeabilize fungal cell membrane, the **azoles** (e.g. ketoconazole, fluconazole, **itraconazole**, **voriconazole**, clotrimazole), which inhibit a P450-dependent enzyme of lanosterol 14- α demethylation thereby decreasing ergosterol synthesis, and the **allylamines** (e.g. **terbinafine**), which inhibit squalene epoxidase, another enzyme involved in ergosterol synthesis. Using a combination of antifungals with different mechanisms of action may have a synergistic effect and may reduce fungal resistance. Voriconazole is the newest antifungal used in birds and seem particularly effective. Endoscopic guided local instillation over fungal plaques or intragranuloma injections of antifungals are also advised depending on the situation. Intratracheal administration of amphotericin B may also be performed. **The use of multiple routes and methods of delivery and antifungals with different mechanisms of action may enhance effectiveness and is still well tolerated in most patients.** Endoscopic debridement of lesions is recommended whenever feasible.



Non-infectious diseases

Birds are very sensitive to smoke inhalation toxicosis and airborne toxins because of their large and efficient pulmonary exchange surface. Toxic fumes frequently include **PTFE** (polytetrafluoroethylene=Teflon) from overheated non-stick cookware but also cigarette smoke, household aerosols (house perfumes, candles, disinfectants), carbon monoxide, ammonia (chickens), and smoke from wild fires.



Signs are most often acute, especially in PTFE inhalation and other toxic fumes, and severe pulmonary hemorrhage and congestion usually ensue. **Oxygen therapy** is the mainstay of treatment and these patients should also be preemptively treated with antibiotics and antifungals. NSAIDs may also be used to decrease inflammation.

Neoplasia of the lower respiratory tract include pulmonary adenocarcinoma, air sac adenocarcinoma, and metastasis of malignant tumors. Pulmonary/bronchogenic carcinoma have been reported in multiple bird species and often manifest as hind limb ataxia/paresis due to local invasion of the tumor into the spine.



Undifferentiated neoplasia of the lungs and air sacs have been recognized in cockatiels and seem to be most often diagnosed in the thoracic inlet occasionally compressing the trachea.



Treatment is limited for respiratory neoplasia, but chemotherapy and radiation therapy may be attempted if diagnosed early on.

Several other non-infectious diseases of clinical significance have been reported in captive birds. **Aspiration pneumonia** may be encountered in baby birds (chronic poor doer) and hospitalized birds being gavage fed. Chronic **pulmonary interstitial fibrosis** has been reported in a number of birds.

A **chronic obstructive respiratory disease is also seen in macaws**, particularly in the blue and gold macaw, housed with feather powder producing species such as cockatoos, cockatiels, and grey parrots (also called the **macaw asthma**). This syndrome is characterized by polycythemia and pulmonary congestion. Lung biopsies can be obtained endoscopically to confirm the diagnosis and assess its severity. It is typically treated with short or long-term bronchodilators. Consequently, **macaws should not be kept in an area with dander-producing parrot species**.

EXTRA-RESPIRATORY DISEASES

Due to the extensive nature of the air sac system in birds in the abdomen and thorax, **any extrarespiratory disease resulting in reduction of air sac volume or**

occlusion of major airways may result in respiratory clinical signs.

In the thoracic inlet, masses of non-respiratory origin may occlude the trachea, syrinx, and interclavicular air sac leading to inspiratory dyspnea, aphonia, or voice changes. These **masses** include thyroid masses (goiter especially in budgerigars, thyroid neoplasia), bony callus from a coracoid fracture, large granulomas, and various neoplasia (see sagittal section of a CT in a parrot with a cystic thymoma below). Cardiomegaly does not typically induce coughing in birds as it does in mammals.



A significant reduction in air space may occur with fluid accumulation in the coelomic cavities of birds such as **ascites** resulting from cardiac, hepatic, or neoplastic diseases, and egg yolk coelomitis. Severe respiratory complications may arise if the fluid/yolk gains access to the inside of respiratory system.

Space-occupying masses are also associated with impairment of abdominal ventilation and are encountered with large tumors (e.g. budgerigars), granuloma, eggs, and organomegaly. Finally, obesity with intracoelomic fat accumulation may impair normal ventilation and induce dyspnea, especially after exercise (see cross-section CT image below of an obese bird, barely any air sac space is seen).

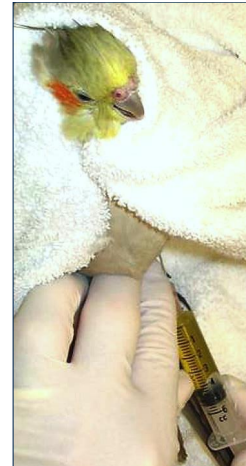


Whole-body imaging is recommended to investigate these disorders such as radiographs or CT-scan. Another thing that can be done is coeloscopy through an interclavicular approach. This air sac can be accessed by

performing a small incision in between the clavicles, reclining the crop on one side, and perforating the interclavicular air sac membrane. This is the approach of choice for tumors or masses within the thoracic inlet.



In terms of treatment of these extrarespiratory disorders, it will depend on the underlying disease. Coelomic fluid should be drained by **abdominocentesis** until dyspnea has improved. Ultrasound can be used to help. Surgical excision or debridement of masses may also be indicated to improve air flow.



The Ten Most Important Facts

1. Birds presented for acute respiratory distress (severe dyspnea, open beak breathing) should be placed in an oxygen cage. They should be preoxygenated prior to doing a restricted physical examination focused on the respiratory system and coelomic palpation.
2. Dyspnea should be characterized into inspiratory, mainly associated with upper respiratory diseases (including trachea), and expiratory/mixed, mainly associated with lower respiratory diseases.
3. The diagnostic workup of avian respiratory diseases include blood work (CBC, biochemistry), whole-body imaging (radiographs, CT) and endoscopy (rhinoscopy, choanoscopy, tracheoscopy, coelioscopy). Coelioscopy can be performed from within the avian air sac system.
4. Oxygen therapy is extremely beneficial in birds, especially with lung parenchymatous diseases. It is mostly accomplished by using an oxygen cage (incubator).
5. Nebulization is frequently used in birds to treat respiratory diseases as it allows locally high concentration of drugs without systemic absorption. Jet nebulizers are frequently used with a particle size of under 3 μm .
6. Avian chlamydiosis is caused by *Chlamydia psittaci* and is a major zoonosis of pet birds. Antibiotics with intracellular distribution such as doxycycline and azithromycin are used for treatment. It is a reportable disease in California. Diagnosis is made by a PCR on a oro-choano-cloacal swab.
7. Infraorbital sinusitis presents with a periorbital swelling and is frequently associated with rhinitis and conjunctivitis. *Mycoplasma gallisepticum* is a major cause of sinusitis in poultry and passerines.
8. Aspergillosis is a common fungal infection of the lungs, air sacs, and sometimes the syrinx. It is diagnosed by imaging and coelioscopy and treated with various antifungals, voriconazole being the preferred drug.
9. Birds are very susceptible to airborne toxins, in particular PTFE and fumes. They cause acute pulmonary congestion and are mainly managed by oxygen therapy.
10. Because of the avian respiratory anatomy, any disease causing a compression of the trachea and syrinx (mass in thoracic inlet, interclavicular airsacculitis) or anything reducing air sac space (ascites, obesity, masses) will lead to respiratory signs.